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Privacy-Preserving Machine Learning for Heart Disease Detection Using Fully Homomorphic Encryption

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Abstract

With the growing adoption of Artificial Intelligence (AI) in sensitive sectors such as healthcare and finance, protecting user privacy during data processing has become paramount. One promising approach is Fully Homomorphic Encryption (FHE), which offers a viable solution by allowing computations to be performed directly on encrypted data, thus safeguarding sensitive information. In this study, we investigate the practical application of the Cheon Kim Kim Song (CKKS) FHE scheme to perform inference with various machine learning models for heart disease detection. We evaluated five models: Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, and a simple Neural Network, across multiple heart disease datasets. Our analysis compares their performance on both standard (plain-text) and encrypted data, using metrics including accuracy, precision, recall, and F1-score. Results demonstrate that encrypted models deliver predictive accuracy comparable to their standard counterparts, confirming the viability of privacypreserving inference with FHE despite the expected increase in computational time. Furthermore, our findings highlight up to 100% consistency between the predictions made on encrypted and plain-text inputs.

Keywords

artificial intelligence, ckks, computing and processing, healthcare, homomorphic encryption, privacy preservation

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***Abstract*—With the growing adoption of Artificial Intelligence (AI) in sensitive sectors such as healthcare and finance, protecting user privacy during data processing has become paramount. One promising approach is Fully Homomorphic Encryption (FHE), which offers a viable solution by allowing computations to be performed directly on encrypted data, thus safeguarding sensitive information. In this study, we investigate the practical application of the Cheon Kim Kim Song (CKKS) FHE scheme to perform inference with various machine learning models for heart disease detection. We evaluated five models: Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, and a simple Neural Network, across multiple heart disease datasets. Our analysis compares their performance on both standard (plain-text) and encrypted data, using metrics including accuracy, precision, recall, and F1-score. Results demonstrate that encrypted models deliver predictive accuracy comparable to their standard counterparts, confirming the viability of privacy-preserving inference with FHE despite the expected increase in computational time. Furthermore, our findings highlight up to 100% consistency between the predictions made on encrypted and plain-text inputs.**

***Index Terms*—Healthcare, Artificial Intelligence, Privacy preservation, Homomorphic Encryption, CKKS.**

I. Introduction

Artificial Intelligence (AI) is transforming sectors such as healthcare and finance through data-driven decision-making. In healthcare, Machine Learning (ML) models support diagnosis, outcome prediction, and clinical decisions. However, these models rely on sensitive patient data (e.g., medical histories, lab results, imaging), raising privacy concerns when processed in third-party or cloud environments. Protecting such data is both an ethical requirement and a legal obligation under regulations such as HIPAA [1], GDPR [2], and PCI DSS [3], which restrict how personal data can be shared and processed.

Conventional protections like anonymization or encryption at rest/in transit still require decryption during processing, creating vulnerability. Fully Homomorphic Encryption (FHE) overcomes this by enabling computation directly on encrypted data, keeping raw data hidden throughout. Although offering strong privacy guarantees, FHE adoption is hindered by high computational cost and limited practical evaluation. This raises a central research question: *Can FHE be applied efficiently to healthcare ML tasks without sacrificing model performance?*

This work addresses the question by benchmarking ML models on encrypted vs. plain-text data using the Cheon–Kim–Kim–Song (CKKS) scheme, designed for approximate arithmetic. We evaluated five classifiers—Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF) and Neural Network (NN)—on multiple heart disease datasets. Results show that encrypted inference yields performance nearly identical to plain-text, demonstrating the feasibility of privacy-preserving ML despite the additional computational cost.

The paper is organized as follows. Section 2 reviews background and related work, Section 3 describes the methodology, Section 4 presents results, and Section 5 concludes with future directions.

II. Background and Related Work

In recent years, integrating privacy-preserving techniques with ML has gained increasing attention, especially in healthcare. As noted in the Introduction, HE enables computation on encrypted data, ensuring privacy [4], [5].

HE schemes can be classified as:

- **Partially Homomorphic Encryption (PHE):** Supports one operation, e.g., multiplication (RSA) or addition (Paillier).
- **Somewhat Homomorphic Encryption (SHE):** Allows limited operations before noise accumulation halts computation.
- **Fully Homomorphic Encryption (FHE):** Supports unlimited operations via bootstrapping, first proposed by Gentry (2009) [6]. Popular FHE schemes include BFV, BGV, FHEW, TFHE, and CKKS.

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Despite its promise, FHE entails high computational cost and lacks native support for non-linear functions (e.g., sigmoid, softmax), requiring polynomial approximations that may reduce ML accuracy.

Several works explored HE for healthcare ML. Han et al. [7] used CKKS for privacy-preserving Naive Bayes inference. Sarkar et al. [8] optimized LR with BFV through efficient matrix multiplication in genomic data. Gogoi and Valan [9] applied Paillier for encrypted inference in chronic kidney disease detection, achieving 98.75% accuracy. Naresh et al. [10] developed HELR using CKKS for heart disease prediction with performance comparable to plain-text models.

Table I summarizes these studies and contrasts them with our approach.

III. APPROACH AND METHODOLOGY

This study aims to benchmark Standard-ML classifiers against their FHE-ML counterparts, with a particular focus on heart disease prediction in the medical domain. Five classifiers were selected for this purpose: LR, SVM, DT, RF, and NN. The following section outlines the experimental framework and methodology used to build and compare these models in both standard and encrypted settings. We begin with a description of the heart disease datasets used, emphasizing their relevance to real-world clinical applications. We then introduce the CKKS encryption scheme used to perform real-valued computations in the encrypted domain and explain how it was integrated into our workflow. Finally, we describe the configurations of each classifier.

A. Datasets

This study uses three heart disease datasets to train and evaluate ML models in both plain-text and encrypted-data. These datasets offer diverse clinical and demographic features relevant to heart disease prediction. As shown in Table II, the Cleveland dataset (303 instances) and the Hungarian dataset (294 instances) were sourced from the UCI Repository. To improve generalization, these datasets were merged, along with others, into a unified dataset with 1,190 instances and 12 harmonized features. Redundant or inconsistent attributes were removed during pre-processing.

TABLE II: Comparison of Used Heart Disease Datasets

Dataset	Source	Instances / Features	Target (Type / Classes)
Cleveland	UCI / Kaggle	303 / 14	Binary: 0 = No, 1 = Yes
Hungarian	UCI / Kaggle	294 / 14	Multi-class (0–4), merged to binary
Combined	Merged Sources	1,190 / 12	Binary: 0 = No, 1 = Yes

B. CKKS Scheme

We adopt the CKKS scheme for its compatibility with real-valued data and practicality in encrypted ML inference. Unlike BFV and BGV, which operate on exact integers, CKKS supports approximate arithmetic on real and complex numbers, making it well-suited for ML workloads where floating-point operations dominate and small errors are tolerable. It encodes vectors of real numbers into polynomials via scaling, enabling encrypted addition, multiplication, and dot products. CKKS is based on the Ring Learning With Errors (RLWE) assumption and supports SIMD operations for efficient batch processing.

Compared to other FHE schemes (Table III), CKKS offers:

- Efficient linear algebra operations
- Native support for real-valued ML tasks
- Polynomial approximation of non-linear functions, enabling NN inference

These features make CKKS the most suitable choice for our heart disease prediction experiments with LR, SVM, DT, RF, and NN models.

TABLE III: Comparison of FHE schemes for machine learning

Feature	**CKKS**	**BFV**	**BGV**	**TFHE**
Arithmetic Type	Approx. real nums	Exact ints	Exact ints	Boolean (bitwise)
Supports real-valued ML	Yes	Needs enc.	Needs enc.	No
Performance (Linear Alg.)	High	Medium	Medium	Low
Efficient Vector Ops (SIMD)	Yes	Yes	Yes	No
Non-linear Support	Poly. approx.	Bitwise methods	Bitwise methods	Native Boolean
Ease in ML Pipelines	High	Low	Low	Very low
Neural Net Support	Yes (approx.)	Not practical	Not practical	No
Post-Quantum Security	Yes (RLWE)	Yes (RLWE)	Yes (RLWE)	Yes (RLWE)

C. FHE-Classifiers

This section outlines the approach and methodology used to implement and compare various ML classifiers: LR, SVM, DT, RF, and NN, in both Standard and FHE environments using plain-text and encrypted-data respectively. All encrypted models were implemented using the CKKS scheme through the TenSEAL library. To ensure a fair comparison, we used nearly identical model architectures for both the Standard and FHE versions, with the only differences being the integration of FHE and the mathematical approximations (such as polynomial or piecewise functions) required to enable encrypted computation. The dataset was split approximately 80/20 into training and testing sets, and the same data partitions were used to train and evaluate both versions of each model. The encrypted implementations were designed to perform inference entirely within the encrypted domain, preserving data confidentiality throughout the computation pipeline.

1) Standard-LR vs. FHE-LR: The FHE-LR model was implemented using the `TenSEAL` library, with encryption conducted under a CKKS context configured with a polynomial modulus degree of 16,384, coefficient moduli tailored for adequate computational depth, and a global scale defined as:

$$\text{scale} = 2^{40}. \tag{1}$$

TABLE I: Comparison of Related Work Based on HE and ML Techniques

Study	HE Technique	ML Technique	Dataset Description
Han et al. [7]	CKKS	Naive Bayes	Breast cancer dataset
Sarkar et al. [8]	BFV	Logistic Regression	Cancer type classification (genomic data)
Gogoi and Valan [9]	Paillier	Logistic Regression	Chronic Kidney Disease dataset
Naresh et al. [10]	CKKS	Logistic Regression	3 heart disease datasets
Our Work	CKKS	LR, SVM, DT, RF, NN	Heart disease datasets (3 datasets, varying sizes)

The encrypted inference pipeline mirrored Standard-LR operations, except all computations were performed directly on ciphertext. Input vectors were encrypted using `ts.ckks_vector`, dot products were computed between encrypted features and plain text weights, an encrypted bias was added, and predictions were generated using a third-degree polynomial approximation of the sigmoid activation function:

$$\sigma(z) \approx 0.5 + 0.197z - 0.004z^3, \tag{2}$$

where $\sigma(z)$ is the approximated sigmoid function used in encrypted inference. The difference between the sigmoid and its approximation is shown in Fig. 1.

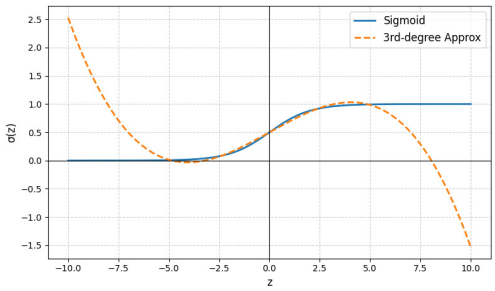


Fig. 1: LR approximation

The encrypted outputs were then decrypted and thresholded at 0.5 for binary classification. In Table IV, we provide a comparison between FHE-LR and Standard-LR.

Unlike many prior works that partially encrypt inference steps, this study implements fully encrypted inference, ensuring that no intermediate values are ever decrypted during computation. The evaluation included not only traditional performance and resource usage metrics, but also the prediction agreement rate between both models to quantify consistency under encryption.

TABLE IV: Comparison of Standard-LR and FHE-LR

Operation	Standard-LR	FHE-LR
Linear Combination	$w \cdot x + b$	Homomorphic computation of $w \cdot x$ with encrypted x, followed by encrypted addition of b
Sigmoid Function	$\frac{1}{1+e^{-z}}$	$0.5 + 0.197z - 0.004z^3$ (polynomial approximation)
Decision Rule	prob $> 0.5 \rightarrow$ class 1	Decrypt, then prob $> 0.5 \rightarrow$ class 1
Encryption Method	None (cleartext)	CKKS homomorphic encryption

This methodology provides a realistic end-to-end view. It captures the computational and performance implications of deploying ML models in FHE.

2) Standard-SVM vs. FHE-SVM: Standard-SVM models were trained using the `scikit-learn` library. For multiclass scenarios (e.g., the Hungarian dataset), both native multiclass models and binary One Vs Rest classifiers were trained to allow consistent comparison. In Table V we present the modifications made to implement FHE in comparison to Standard-SVM.

The FHE-SVM model was implemented using the `TenSEAL` library. The encryption context was configured with a polynomial modulus degree of 8192, coefficient modulus sizes of $\{40, 30, 30, 40\}$, and a global scale of 2^{30} to maintain precision. The encrypted inference pipeline consisted of the following steps:

1) Extraction of model parameters (weights and bias) from the trained SVM.
2) Encryption of the entire test dataset using the `CKKSVector`.
3) Computation of dot products between encrypted input vectors and plain text weights.
4) Encrypted bias addition.
5) Decryption and binarization of results to obtain final predictions.

This experiment demonstrates a complete privacy-preserving SVM pipeline that operates fully within the encrypted domain.

TABLE V: Comparison of Standard-SVM and FHE-SVM

Operation	Standard-SVM	FHE-SVM
Kernel Computation	Linear Kernel: $x \cdot x'$	Encrypted Linear Kernel: $\text{enc}(x) \cdot x'$
Decision Function	$\sum(w_i \cdot x_i) + b$	Encrypted dot product $\text{enc}(x) \cdot w$·
Classification Rule	sign(decision_function(x))	Decrypt, then sign(decision_value)

3) Standard-DT vs FHE-DT: $\sum(w_i \cdot x_i) + b$

This section presents the architecture, implementation, and mathematical basis for the benchmark comparison between a Standard-DT and its privacy-preserving counterpart operating over encrypted-data using FHE with the CKKS scheme. The primary objective is to evaluate the trade-offs between privacy, performance, and resource utilization. In Table VI, summarizes the main differences between Standard-DT and FHE-DT.

We use the `DecisionTreeClassifier` from Scikit-learn with a fixed depth (typically `max_depth = 3`) for consistency. The tree is trained on normalized features and the

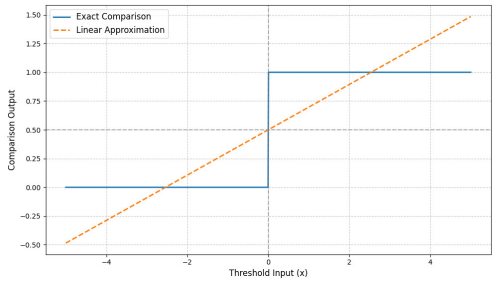


Fig. 2: DT approximation

model produces deterministic splits by comparing each input feature with a threshold at internal nodes. A decision node in the standard tree computes:

$$f_i(x) = \begin{cases} \text{go to left child} & \text{if } x_j \leq t_j, \\ \text{go to right child} & \text{if } x_j > t_j, \end{cases}$$

where x_j is the jth feature and t_j is the threshold for that node. This operation uses a precise, non-linear comparison, and the traversal is exact.

In the FHE variant, we leverage `TenSEAL` to implement CKKS-based encryption, enabling computation on encrypted floating-point vectors. The key challenge is that CKKS supports approximate arithmetic but not conditional branches or comparisons. To address this limitation and still enable decision-making within the encrypted domain, we adapt the traditional tree traversal process using FHE-compatible techniques. We proceed as follows:

1) Train the same decision tree structure on plain-text
2) Encrypt each feature of every test sample using CKKS
3) Simulate decision logic with polynomial approximations
4) Traverse the tree using FHE-compatible operations

we chose parameters as follows:

- Polynomial modulus degree: 8192
- Coeff modulus sizes: `[60, 40, 40, 60]`
- Global scale: 2^{40}
- Security Level: $\sim$128 bits

Since CKKS does not support comparisons such as $x > t$, we use a polynomial approximation to simulate binary decisions. As shown in Fig. 2, we approximate the step function using a linearized sigmoid function:

$$\tilde{f}(x) = 0.5 + 0.2(x - t), \tag{3}$$

where x is the feature (in encrypted form) and t is the decision threshold (in plain-text). If $\tilde{f}(x) > 0.5$, we move to the right; otherwise, we move to the left. For the FHE imprecision, we inject small Gaussian noise into this result. At leaf nodes, the stored prediction is the class probability. We encrypt this scalar and return it to the client, who decrypts it and applies thresholding:

$$\text{Prediction} = \begin{cases} 1 & \text{if decrypted output } > 0.5, \\ 0 & \text{otherwise.} \end{cases}$$

To manage memory limitations, we process encrypted test data in batches of 50 samples. Each test feature is encrypted individually as:

$$x_j^{(i)} \rightarrow \texttt{CKKSVector}([x_j^{(i)}]). \tag{4}$$

The entire test set is encrypted to ensure that the benchmark reflects real-world privacy-preserving inference.

TABLE VI: Comparison of Standard-DT vs. FHE-DT

Aspect	Standard-DT	FHE-DT
Comparison	Exact: $x > t$	Approximate: $0.5 + 0.2(x - t)$
Precision	Full (float64)	Approximate (CKKS, limited by scale)
Tree Traversal	Deterministic	Approximate with injected noise
Branching	Conditional	via polynomial thresholds
Output	Hard classification	Soft probability, then decrypted

4) Standard-RF vs FHE-RF: Standard RFs rely on `if-else` branching, but CKKS only supports vectorized approximate arithmetic and does not allow direct comparisons. Thus, each tree is reduced to a *decision stump* (depth-1) and branching is replaced with a polynomial approximations.
CKKS Parameters: Polynomial modulus degree: 8192, Coefficient modulus sizes: [60,40,40,60], Global scale: 2^{40}.
Decision Stump S_i: Feature f_i, threshold θ_i, predictions $p_{\text{left},i}, p_{\text{right},i}$.

Since comparisons are not supported, we compute a soft decision:

$$c = \sigma_{\text{approx}}(\theta - x). \tag{5}$$

Prediction for stump i:

$$\hat{y}_i = p_{\text{left},i} \cdot c + p_{\text{right},i} \cdot (1 - c), \tag{6}$$

Final output:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^{N} \hat{y}_i, \tag{7}$$

with binarization at 0.5. We proceed as follows:

1) Encrypt dataset with CKKS,
2) Evaluate with polynomial logic,
3) Decrypt predictions.

In summary, FHE-RF achieves fully encrypted inference without mid-pipeline decryption. Decision stumps and polynomial approximations (sigmoid/tanh) ensure FHE compatibility while balancing simplicity and interpretability.

5) Standard-NN vs FHE-NN: Standard-NN was implemented using PyTorch, comprising an input layer, a hidden layer with 8 ReLU-activated neurons, and a sigmoid output layer.

The forward pass of the neural network is described by Equations (12)–(15). First, the input vector x is linearly transformed using the weight matrix W_1 and bias b_1, resulting in the pre-activation vector z_1 (Equation (12)). Next, the ReLU

activation function is applied element-wise to z_1, producing the hidden layer activation a_1 (Equation (13)). This activation is then passed through another linear transformation with weights W_2 and bias b_2, yielding the output pre-activation z_2 (Equation (14)). Finally, the sigmoid function σ is applied to z_2, generating the network's prediction $\hat{y}$ (Equation (15)). The equations are expressed as follows:

$$z_1 = W_1 x + b_1, \tag{8}$$

$$a_1 = \text{ReLU}(z_1), \tag{9}$$

$$z_2 = W_2 a_1 + b_2, \tag{10}$$

$$\hat{y} = \sigma(z_2), \tag{11}$$

where $\text{ReLU}(x) = \max(0, x)$, and $\sigma(x) = \frac{1}{1+e^{-x}}$. The loss function used is binary cross-entropy:

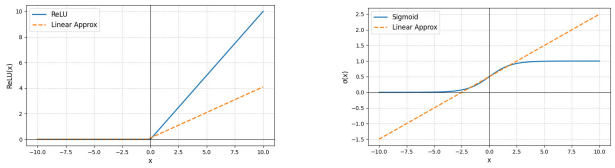
$$L(y, \hat{y}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y}). \tag{12}$$

In contrast, the FHE-NN built using the TenSEAL library and CKKS encryption replaces nonlinearities with low-degree polynomial approximations to comply with encryption constraints. Specifically, ReLU is approximated as shown in Fig. 3a:

$$\text{ReLU}_{\text{approx}}(x) = 0.1 + 0.4x, \tag{13}$$

and the sigmoid is approximated as shown in Fig. 3b:

$$\sigma_{\text{approx}}(x) = 0.5 + 0.2x. \tag{14}$$



(a) ReLU function vs linear approximation

(b) Sigmoid function vs linear approximation

Fig. 3: Neural network approximations

allowing all operations to be performed solely via additions and multiplications on encrypted vectors.

The encrypted forward pass computes:

$$z_{1,i} = \sum_{j=1}^{n} w_{ij}^{(1)} x_j + b_i^{(1)}, \tag{15}$$

followed by approximate linear activation and encrypted summations through the hidden and output layers:

$$z_2 = \sum_{i=1}^{h} w_i^{(2)} a_i + b^{(2)}, \quad \hat{y}_{\text{enc}} = \sigma_{\text{approx}}(z_2). \tag{16}$$

All operations are performed in the encrypted domain using the CKKS scheme, which supports approximate arithmetic over real numbers. Parameters were tuned for precision and circuit depth, with a polynomial modulus degree set to 16,384 and the coefficient modulus chain to $[60, 40, 40, 40, 40, 60]$.

IV. Results and Discussion

This section provides a thorough evaluation of the previously introduced classifiers, comparing their performance under plain-text conditions with their homomorphically encrypted counterparts. The analysis is divided into two main components: first, we assess classification performance through standard performance metrics and model agreement. Second, we examine the system-level overhead imposed by encrypted inference in terms of execution time.

A. Classification Performance: Accuracy, Precision, Recall, F1 Score, and Model Agreement

FHE preserved predictive integrity across all classifiers, with minor variations by model and dataset.

LR achieved perfect consistency (100% agreement) in all datasets. SVM maintained full agreement on the Cleveland and Combined datasets but dropped slightly (94.38%) on the Hungarian set, with modest declines in accuracy and recall. DT models showed strong alignment (>97% agreement) and only slight reductions in precision or F1. RF preserved perfect recall across datasets and 100% agreement on two of them, with only minor metric drops in the Combined dataset. NN exhibited the greatest degradation, with agreement between 88–92% and accuracy falling 3–5%, yet overall predictive behavior remained stable.

Tables VII–IX summarize the detailed metrics. Overall, FHE-models maintained high agreement with their plain-text versions, confirming that secure inference is feasible with only marginal performance loss.

TABLE VII: Classification Performance for the *"Cleveland"* Dataset

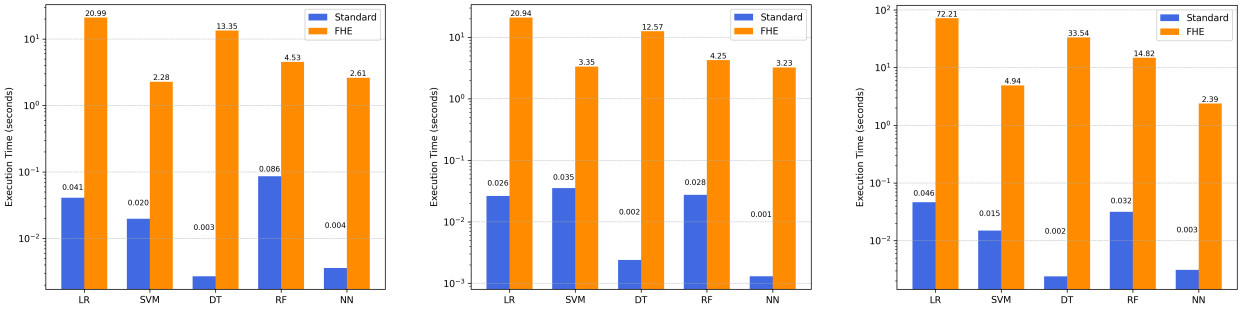
Model	**Accuracy**		**Precision**		**Recall**		**F1**		**Agreement**
–	Std.	FHE	Std.	FHE	Std.	FHE	Std.	FHE	–
LR	0.8525	0.8525	0.8710	0.8710	0.8438	0.8438	0.8571	0.8571	100%
SVM	0.8033	0.8033	0.7561	0.7561	0.9394	0.9394	0.8378	0.8378	100%
DT	0.8197	0.8033	0.8387	0.8125	0.8125	0.8125	0.8254	0.8125	98.61%
RF	0.7560	0.7560	0.7560	0.7560	1	1	0.8611	0.8611	100%
NN	0.8525	0.8197	0.9259	0.8387	0.7812	0.8125	0.8475	0.8254	90.16%

TABLE VIII: Classification Performance the *"Hungarian"* Dataset

Model	**Accuracy**		**Precision**		**Recall**		**F1**		**Agreement**
–	Std.	FHE	Std.	FHE	Std.	FHE	Std.	FHE	–
LR	0.7966	0.7966	0.8000	0.8000	0.5714	0.5714	0.6667	0.6667	100%
SVM	0.8764	0.8427	0.8182	0.8214	0.8438	0.7188	0.8308	0.7667	94.38
DT	0.7966	0.7966	0.8462	0.8000	0.5238	0.5714	0.6471	0.6667	97.16%
RF	0.6949	0.6949	0.6949	0.6949	1	1	0.8200	0.8200	100%
NN	0.8475	0.8305	0.9286	0.9231	0.6190	0.5714	0.7429	0.7059	98.30%

B. System Overhead: Inference Time

FHE introduces substantial computational overhead for all models. For example, Logistic Regression under FHE shows a slowdown of roughly 951× compared to standard inference, while SVM experiences an average slowdown of about 82×. Decision Trees and Random Forests also see significant increases, with RF slowing down approximately 163×. NN under



(a) Cleveland dataset (b) Hungarian dataset (c) Combined dataset

Fig. 4: Inference time comparison between standard and FHE-encrypted models

TABLE IX: Classification Performance for the *"Combined"* Dataset

Model	**Accuracy**		**Precision**		**Recall**		**F1**		**Agreement**
–	Std.	FHE	Std.	FHE	Std.	FHE	Std.	FHE	–
LR	0.8613	0.8613	0.8712	0.8712	0.8779	0.8779	0.8745	0.8745	100%
SVM	0.8361	0.8361	0.8425	0.8425	0.8492	0.8492	0.8458	0.8458	100%
DT	0.8277	0.8277	0.8571	0.8309	0.8244	0.8626	0.8405	0.8464	98.24%
RF	0.8032	0.7868	0.8000	0.7868	1	1	0.8888	0.8807	99.06%
NN	0.8193	0.7689	0.8284	0.7159	0.8473	0.9618	0.8377	0.8208	82.35%

FHE show a similar trend, with inference times increasing by orders of magnitude.

Despite these overheads, FHE CKKS remains practical for offline or semi-interactive applications in privacy-sensitive settings. Optimization techniques such as quantization, efficient polynomial approximations, or hardware acceleration are essential to approach real-time performance.

V. Conclusion and Future Work

This study benchmarked FHE on LR, SVM, DT, RF, and NN for heart disease prediction. The results show that FHE models closely match their plain-text counterparts, confirming both correctness and privacy-preserving potential. The main drawback is computation: Inference under FHE is slower by several orders of magnitude, especially for complex models, which limits real-time use. This underscores the trade-off between privacy and efficiency. FHE is already suitable for privacy-critical domains like healthcare, but a wider adoption will require model optimization, hardware acceleration, or hybrid methods to reduce overhead while preserving security.

References

[1] U.S. Department of Health and Human Services, "Health information privacy," 2017, accessed: 2025-06-05. [Online]. Available: https://www.hhs.gov/hipaa/index.html

[2] GDPR.ORG, "Eu general data protection regulation (gdpr) portal: Site overview," 2017, accessed: 2025-06-05. [Online]. Available: http://www.eugdpr.org/

[3] Payment Card Industry, "Payment card industry data security standard (pci dss)," 2016, version 3.2.

[4] T. V. T. Doan, M.-L. Messai, G. Gavin, and J. Darmont, "A survey on implementations of homomorphic encryption schemes," *The Journal of Supercomputing*, vol. 79, no. 13, pp. 15 098–15 139, 2023. [Online]. Available: https://doi.org/10.1007/s11227-023-05233-z

[5] K. Munjal and R. Bhatia, "A systematic review of homomorphic encryption and its contributions in healthcare industry," *Complex & Intelligent Systems*, vol. 9, no. 4, pp. 3759–3786, 2023. [Online]. Available: https://doi.org/10.1007/s40747-022-00756-z

[6] C. Gentry, *A fully homomorphic encryption scheme*. Stanford university, 2009.

[7] B. Han, Y. Kim, J. Choi, H. Shin, and Y. Lee, "Fully homomorphic privacy-preserving naive bayes machine learning and classification," in *Proceedings of the 11th Workshop on Encrypted Computing & Applied Homomorphic Cryptography*, ser. WAHC '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 91–102. [Online]. Available: https://doi.org/10.1145/3605759.3625262

[8] E. Sarkar, E. Chielle, G. Gursoy, L. Chen, M. Gerstein, and M. Maniatakos, "Privacy-preserving cancer type prediction with homomorphic encryption," *Scientific Reports*, vol. 13, no. 1, p. 1661, 2023, published on 2023-01-30. [Online]. Available: https://doi.org/10.1038/s41598-023-28481-8

[9] P. Gogoi and J. A. Valan, "Application of homomorphic encryption in machine learning based chronic kidney disease prediction," in *2024 2nd World Conference on Communication Computing (WCONF)*, 2024, pp. 1–6.

[10] V. S. Naresh and S. Reddi, "Exploring the future of privacy-preserving heart disease prediction: a fully homomorphic encryption-driven logistic regression approach," *Journal of Big Data*, vol. 12, no. 1, p. 52, 2025. [Online]. Available: https://doi.org/10.1186/s40537-025-01098-6